

CPRE/CYBSC-5360_FALL-2025 PROJECT-3

Questions

1. (1 point) Please find all the apps installed and recently used on Betty's phone.

When opening up the USERDATA for BettyNote2Black in Encase, I was able to find the folders app-lib and app. When I looked into these I found all the apps on Betty's phone.

There are 34 apps in the app-list folder:

☐ 1	com.amazon.dee.app-1	app-1
☐ 2	com.android.chrome-1	chrome-1
☐ 3	com.android.vending-1	vending-1
☐ 4	com.dropbox.android-1	android-1
☐ 5	com.ENP.therings.kr.googleplay-1	googleplay-1
☐ 6	com.google.android.apps.access.wi...	consumer-1
☐ 7	com.google.android.apps.books-1	books-1
☐ 8	com.google.android.apps.docs-1	docs-1
☐ 9	com.google.android.apps.magazine...	magazines-1
☐ 10	com.google.android.apps.maps-1	maps-1
☐ 11	com.google.android.apps.plus-1	plus-1
☐ 12	com.google.android.gm-1	gm-1
☐ 13	com.google.android.gms-1	gms-1
☐ 14	com.google.android.googlequickse...	googlequicksearchbox-2
☐ 15	com.google.android.marvin.talkbac...	talkback-1
☐ 16	com.google.android.play.games-1	games-1
☐ 17	com.google.android.talk-1	talk-1
☐ 18	com.google.android.tts-1	tts-1
☐ 19	com.google.android.videos-1	videos-1
☐ 20	com.google.android.youtube-1	youtube-1
☐ 21	com.hp.android.printservice-2	printservice-2
☐ 22	com.kingoapp.superbattery-1	superbattery-1
☐ 23	com.lguplus.rms-1	rms-1
☐ 24	com.lguplus.usimsvcm-1	usimsvcm-1
☐ 25	com.nest.android-1	android-1
☐ 26	com.policydm-1	policydm-1
☐ 27	com.sec.android.app.samsungapps-1	samsungapps-1
☐ 28	com.sec.android.fwupgrade-1	fwupgrade-1
☐ 29	com.sec.app.samsungprintservice-1	samsungprintservice-1
☐ 30	com.sec.pcw-1	pcw-1
☐ 31	com.sec.penup-1	penup-1
☐ 32	com.sec.spp.push-1	push-1
☐ 33	com.uplus.onphone-1	onphone-1
☐ 34	com.xiaomi.hm.health-1	health-1

And there are 36 apps in the app folder:

☐	1		com.ENP.therings.kr.googleplay-1.a...	apk	98,603,626	Application
☐	2		com.dropbox.android-1.apk	apk	54,733,737	Application
☐	3		com.google.android.googlequickse...	apk	50,339,804	Application
☐	4		com.android.chrome-1.apk	apk	47,994,996	Application
☐	5		com.amazon.dee.app-1.apk	apk	45,345,831	Application
☐	6		com.nest.android-1.apk	apk	41,567,399	Application
☐	7		com.uplus.onphone-1.apk	apk	39,303,769	Application
☐	8		com.google.android.gms-1.apk	apk	33,936,896	Application
☐	9		com.sec.android.fwupgrade-1.apk	apk	31,993,039	Application
☐	10		com.google.android.talk-1.apk	apk	30,765,615	Application
☐	11		com.google.android.apps.maps-1.a...	apk	29,618,884	Application
☐	12		com.google.android.apps.access.wi...	apk	29,597,483	Application
☐	13		com.google.android.youtube-1.apk	apk	22,649,182	Application
☐	14		com.google.android.apps.docs-1.apk	apk	21,365,411	Application
☐	15		com.sec.penup-1.apk	apk	21,260,517	Application
☐	16		com.xiaomi.hm.health-1.apk	apk	20,747,484	Application
☐	17		com.lguplus.usimsvcm-1.apk	apk	20,486,198	Application
☐	18		com.google.android.gm-1.apk	apk	19,735,239	Application
☐	19		com.google.android.apps.plus-1.apk	apk	19,255,143	Application
☐	20		com.android.vending-1.apk	apk	18,538,635	Application
☐	21		com.sec.android.app.samsungapps...	apk	17,394,802	Application
☐	22		com.sec.app.samsungprintservice...	apk	16,441,277	Application
☐	23		com.google.android.tts-1.apk	apk	14,852,916	Application
☐	24		com.google.android.play.games-1.a...	apk	14,168,414	Application
☐	25		com.google.android.videos-1.apk	apk	13,793,180	Application
☐	26		com.hp.android.printservice-2.apk	apk	13,745,628	Application
☐	27		com.google.android.apps.magazine...	apk	12,202,255	Application
☐	28		com.google.android.apps.books-1....	apk	11,345,315	Application
☐	29		com.google.android.marvin.talkbac...	apk	7,550,615	Application
☐	30		com.lguplus.rms-1.apk	apk	3,517,958	Application
☐	31		com.kingoapp.superbattery-1.apk	apk	2,828,268	Application
☐	32		com.sec.spp.push-1.apk	apk	1,588,893	Application
☐	33		com.sec.pcw-1.apk	apk	918,253	Application
☐	34		com.policydm-1.apk	apk	870,777	Application
☐	35		FFFFFFFFF000000000000000000000...	dr...	19,221	None
☐	36		mcRegistry		4,096	Folder

2. (1 point) Please find the browsing history of websites and videos.

Based on Q1, Betty's phone mainly used the Chrome browser. After searching for the keyword "History" in filenames, I was able to locate the file. When opening this file up in DB Browser for SQLite, I got the following websites from the "urls" table:

	id	url	title
	File...	Filter	Filter
1	1	https://www.google.co.kr/search?q=gmail&oq=gmail&aqs=chrome..69157j0j5j0.2043j0j4&client=ms-android-lgu...	gmail - Google 검색
2	2	https://www.google.com/gmail/	
3	3	https://mail.google.com/mail/	
4	4	https://accounts.google.com/ServiceLogin?service=mail&passive=true&continue=https://mail.google.com/mail/...	
5	5	https://mail.google.com/accounts/SetOSID?...	
6	6	https://mail.google.com/mail/?pli=1&auth=6QQMmrjG4qdYUwKAMoKJ8sOo7wz-d-5jtbJCNIm19aucetqEuvSLJ3gTlY1WlWeyF4mdw. #	
7	7	https://mail.google.com/mail/?pli=1#	
8	8	https://mail.google.com/mail/x/1i69dln4su9dj-/?pli=1&f=1#	
9	9	https://mail.google.com/mail/mu/#	
10	10	https://mail.google.com/mail/mu/mp/576/#	Gmail
11	11	https://mail.google.com/mail/mu/mp/576/#t1/priority/%5Esmartlabel_personal	
12	12	https://mail.google.com/mail/mu/mp/576/#cv/priority/%5Esmartlabel_personal/15d31f384ba73407	
13	13	https://www.amazon.com/Bach-Complete-Organ-Johann-Sebastian/dp/B001R3YJS8	Bach: Complete Organ Works

To find the video history, I looked inside the chrome data and found a log file in the NTPSnippets. This shows some amount of logs for youtube search history:

000003.log

```
00000Efa5.....Efa5.....Efa5.....Efa5.....Efa5.....K-WjÖ.....0x742
001212a85aa5b8caa4·0x7422a85aa5b8caa4·0Blood from young animals can revitalise old ones·;·IT WAS one of the oddest experime
00242nts in the history of dentistry. In the early 1950s a researcher called Benjamin Kamrin was looking into the causes of to
00363oth ... "shhttps://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRVpNkYsXXMUBl3CtWjD5hQxDrhbmVhKMUbsbZ2wYZMUzm7XYMiPcQXy_E7
00484FMI (ÄÖK·Ü·ÖÄ'·8diä··=·î·>B)· ·~https://www.economist.com/news/science-and-technology/21724967-might-be-true-people-too-bl
00605ood-young-animals-can-revitalise-old· The EconomistH·P·XZñšê·Ü· ··0x9baae212a796b6268· ·0x9baae212a796b626·eCRISPRâ€Cas
00726encoding of a digital movie into the genomes of a population of living bacteria : Nature·c·DNA is an excellent medium for
00847 archiving data. Recent efforts have illustrated the potential for information storage in DNA using synthesized oligonucl
00968eotides ... "shhttps://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRGs6Xoo816e49pFFOCMC·XQZSrspRrjnK3PPy5SCZMIVTdj6pjlhIK
01089bVb49zc(êâpAU··0E¥-Z·B··=·>B5 +https://www.nature.com/articles/nature23017·NatureH·P·XZñšê·Ü· ··0xfdbbc939706a39fd·
01210·0xfdbbc939706a39fd·0x00003000b27e7d02·6Kesha - Woman (Official Video) ft. The Dap-Kings Horns·S·New music from Kesha! G
01331et "Woman" off her upcoming album 'Rainbow' on iTunes: http://smarturl.it/iWoman?IQid=yt Stream on Apple Music: ... "shhttp
01452s://encrypted-tbn1.gstatic.com/images?q=tbn:ANd9GcS15Ke2AN-w3_Mg8AR9SdXcQFvFXhwQEsw51rhWrN2PkubluE8s5yx_zWd-7cI (Äö8;ÿÖ·0
01573Äq6Ää·=üê >B6 +https://www.youtube.com/watch?v=lXyA4MXKIKo·YouTubeH·P·XZñšê·Ü· ··0xabb4b4742257d6ea8i· ·0xabb4b4742257d6
01694ea8·0x00003000b27a5b90·Lonzo Ball UNREAL Highlights vs 76ers (2017.07.12) Summer League - 36 Pts, 11 Ast, 8 Reb, EPIC!·
01815'·Become a monster. THE WORLD'S MOST ELITE JUMP PROGRAM: http://goo.gl/RctOlo Take your ball handling to the next level w
01936ith the BALL HOG GLOVES ... "shhttps://encrypted-tbn1.gstatic.com/images?q=tbn:ANd9GcQILMJ7j7Q3i3KSIMi-AcVubNeFdHrnZJf5ikje-
02057T95maq7mwk8tc13jvOxxqhs (Äö·s8·0ÄipBBB·=·pJ >B6 +https://www.youtube.com/watch?v=RNJUmxK40Y·YouTubeH·P·XZñšê·Ü· ··0x58
02178571ef85af2c873· ·0x58571ef85af2c873·NAmerican tourist killed in Greece was fatally beaten over a selfie, police say·'·Am
02299erican Bakari Henderson was killed by a group of as many as 15 men because of an argument over a selfie at a bar on a Gre
02420ek island, police said."shhttps://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSwVpz1fNSAvio-oqiBwpl7TEnjGu5uc9-cr-A9olQG
02541IhW8fW2Bh84BCmViqw4 (ê'ÿž0Ö·0E8-ê|B·=·°·>B· ·yhttp://www.msn.com/en-us/news/us/american-tourist-killed-in-greece-was-fat
02662ally-beaten-over-a-selfie-police-say/ar-BBEfpKO·MSN.comH·P·XZñšê·Ü· ··0x058eb627c3f048ffÿ· ·0x058eb627c3f048ff·Killer B
02783undle X·-·X marks the spot with the 10th edition of our famous Killer Bundle! With 9 out of 10 new-to-bundle Steam games,
02904 you definitely won't want to miss this."shhttps://encrypted-tbn3.gstatic.com/images?q=tbn:ANd9GcSnZXQTlDqiWy_H6r5PIWp6IvJ
03025lqKPBvinmN1XJi9X8Op3AZrez2EwU_q9qxm8 (êi·xüÖ·0EÖÄ·B·=·\·>BE Shhttps://www.bundlestars.com/en/bundle/killer-bundle-x·Bun
03146dle StarsH·P·XZñšê·Ü· ··0x4147ffafee439468ÿ· ·0x4147ffafee439468·0x00003000b2774dbd·>'Batman' Director Matt Reeves Has D
03267ropped Ben Affleck's Script·[The 'War for the Planet of the Apes' helmer is "starting again" and promised a "new story.""
03388shhttps://encrypted-tbn3.gstatic.com/images?q=tbn:ANd9GcQtn6dWYrZ16Jxt3MTTzL-2K759EGC-l3VaiVttusq9wduoHsiMDnX9ConGyK4 (ê'@i
03509··0EÖÄ·-B·=·î·>Bü· phhttp://www.hollywoodreporter.com/heat-vision/batman-director-matt-reeves-has-dropped-ben-afflecks-
03630script-1020694·Hollywood Reporter·thhttp://www.hollywoodreporter.com/amp/heat-vision/batman-director-matt-reeves-has-drop
03751ped-ben-afflecks-script-1020694H·P·XZñšê·Ü· ··0x96280579062b29cb· ·0x96280579062b29cb·YouTube·c·Sorry for the interrupt
03872ion·We have been receiving a large volume of requests from your network·To continue with your YouTube experience, please
```

3. (2 points) Please find the chat history and emails.

After looking inside the chrome folder, I found a file named mailstore.bettyhallym@gmail.com.db. After opening it, I found the below email conversations and messages:

Database Structure Browse Data Edit Pragas Execute SQL				
Table: conversations				
_id	jeryl	subject	snippet	
Filter	Filter	Filter	Filter	
1...	0	Activate your Mi Account today	Dear user, Thank you for creating a Mi Account. Please click the link below t...	
2...	0	Join the Nest home of simon	You have an open invitation. simon wants to give you access to the Nest produ...	
3...	0	Hallym님, 새 Google 계정을 최대한 활용해보세요.	Hallym님, 안녕하세요. Gmail을 사용해 주셔서 감사합니다. 다음은 Gmail 활용도를 높이기 위한 몇 가지 도움말입니다. 시간을 ...	

Database Structure Browse Data Edit Pragas Execute SQL				
Table: messages				
fromAddress	toAddresses	replyToAddresses	subject	
Filter	Filter	Filter	Filter	
1 "Xiaomi Corporation" <noreply-account@xiaomi.com>	" " <bettyhallym@gmail.com>		Activate your Mi Account today	
2 "Nest Invitation" <account@nest.com>	" " <bettyhallym@gmail.com>	"Auto-Away" <do-not-reply@nest.com>	Join the Nest home of simon	
3 "Google 커뮤니티팀" <googlecommunityteam-...	" " <bettyhallym@gmail.com>	"Google 커뮤니티팀" <googlecommunityteam-...	Hallym님, 새 Google 계정을 최대한 활용해보세요.	

I know that the app used for chats was google hangouts but was unable to find any files relating to it.

4. (1 point) Please find the history of usage between smartphone and wearable device.

After filtering for Xiaomi apps by using the keyword “mi”, I was able to come across a few different db files, I looked through mistat.db and found the below:

New Database

Open Database

Write Changes

Revert Changes

Undo

Open Project

Save Project

Attach Database

Close Database

Database Structure

Browse Data

Edit Pragmas

Execute SQL

Table: mistat_event

Filter in any column

	id	category	ts	key	value	type	extra
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	3754	Event	1500276002646	Dashboard_ViewNum	1	event	
2	3755	mistat_session	1500276046304		1500275820512,1500276016276		WIFI
3	3756	mistat_pv	1500276046323		.device.HMMiLiSettingActivity,.activ...		
4	3757	Event	1500276104636	Dashboard_StatusBar	1	event	{"Property":"BleOff"}
5	3758	Event	1500276104636	BleOff	1	event	
6	3759	Event	1500280274725	Dashboard_ConnectionResults	1	event	{"Property":"Fail"}

The above shows the events between betty's smartphone and the wearable device since the file was found on betty's phone.

I went into device details for the wearable logs and found the following:

[illegible]

5. (1 point) Please find the timestamp when Amazon Echo is called by voice command “Wake up”.

After unzipping the 005-Amazon-Echo-Alexa-Web-Scrape file, I was able to go inside and look at all the files one at a time. Towards the end of the folder I found a file called “history second page.png” which contained the following details:

alexa	Today at 3:06 PM on simon's Echo	>
turn on tv	Today at 3:01 PM on simon's Echo	>
alexa	Today at 3:01 PM on simon's Echo	>
alexa stop	Today at 2:45 PM on simon's Echo	>
wake up	Today at 2:45 PM on simon's Echo	>
alexa	Today at 2:45 PM on simon's Echo	>
Text not available. Click to play recording.	Today at 2:31 PM on simon's Echo	>
Text not available. Click to play recording.	Today at 12:07 PM on simon's Echo	>
alexa kinda patton a.m. next monday	Today at 11:57 AM on simon's Echo	>
connect to my device	Yesterday at 8:24 PM on simon's Echo	>
alexa	Yesterday at 8:24 PM on simon's Echo	>

As we can see from the above entries, the Amazon Echo is called by voice command “Wake up” at exactly 2:45 PM.

6. (1 point) Please find the IP address from Google OnHub diagnostic-report.

```
D3A-Q2Q-Q8B", "networkConfig": { "local_network": { "ip_address": "192.168.86.1", "netmask": "255.255.255.0", "nwireless_network": { "ssid": "HOME", "primary_psk_enabled": true, "channel_2400mhz": { "number": 6, "ht_capabilities": "HT_20", "vht_width": "WIDTH_20_OR_40_MHZ", "vht_center_frequency_0": 6, "vht_center_frequency_1": 0 }, "channel_5000mhz": { "number": 36, "ht_capabilities": "HT_40_PLUS", "vht_width": "WIDTH_80_MHZ", "vht_center_frequency_0": 42, "vht_center_frequency_1": 0 }, "ndhcp": { "pool_begin": 20, "pool_end": 250 }, "nwan": { "connection_type": "CONNECTION_TYPE_DHCP", "nenable_anonymous_metrics_collection": true, "nenable_upnp": true, "nenable_background_data_collection": true, "nenable_traffic_acceleration": true, "nversion": 6, "nguest_network": { "enabled": true, "ssid": "HOME-guest", "psk": "*****", "welcome_mat": { "enabled": true, "title": "", "introduction_text": "", "photo_url": "" }, "nmesh_network": { "ssid": "4E2F1AD8", "psk": "*****", "nmeasured_wan_speed": { "upstream_bytes_per_second": 5636882, "downstream_bytes_per_second": 11153857 }, "fileLengths": [ { "path": "/mnt/stateful_partition/var/log/messages.3", "length": 1926304 }, { "path": "/mnt/stateful_partition/var/log/messages.1", "length": 1849751 }, { "path": "/mnt/stateful_partition/var/log/messages.4", "length": 1831177 }, { "path": "/mnt/stateful_partition/var/log/messages.2", "length": 2181958 }, { "path": "/mnt/stateful_partition/var/log", "length": 7980426 }, { "path": "/mnt/stateful_partition/var/log/update_engine/update_engine.19700101-000009", "length": 1113715 }, { "path": "/mnt/stateful_partition/var/log/update_engine", "length": 1113715 } ], "wanInfo": { "enable_prioritized_device": false, "ndisable_lan": false, "nwan_interface": { "wan0": { "ndhcp_lease": { "mac_address": "2016d8000001", "ip_address": "192.168.86.29", "ndhcp_lease": { "mac_address": "109266000002", "ip_address": "192.168.86.28", "ndhcp_lease": { "mac_address": "d052a8000003", "ip_address": "192.168.86.27", "ndhcp_lease": { "mac_address": "b827eb000004", "ip_address": "192.168.86.25", "ndhcp_lease": { "mac_address": "50f520000005", "ip_address": "192.168.86.26", "ndhcp_lease": { "mac_address": "1caf05000006", "ip_address": "192.168.86.24", "ndhcp_lease": { "mac_address": "18b430000007", "ip_address": "192.168.86.22", "ndhcp_lease": { "mac_address": "a002dc000008", "ip_address": "192.168.86.21", "nwan_ipv4_address": "192.168.165.9", "nupstream_name_server": "210.115.225.11", "nenable_group_setup_network": false, "ndisable_wireless_lan": false, "nenable_station_mode": false, "nwan_ipv4_gateway": "192.168.165.1", "nis_channel_149_allowed": true, "nlan_link_local_ipv6_address":
```

After cleaning up the above onhubdump report, I got the following details:

Clean Interface IP Addresses	Clean Active Device IPs
127.0.0.1	192.168.86.21
192.168.86.1	192.168.86.22
192.168.87.1	192.168.86.24
192.168.165.9	192.168.86.25
	192.168.86.26
	192.168.86.27
	192.168.86.28
	192.168.86.29

Another screenshot with cleaned up IP Addresses:

```
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.21\n2017-07-17T07:28:46.165362+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.22\n2017-07-17T07:28:46.171485+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.24\n2017-07-17T07:28:46.174667+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.25\n2017-07-17T07:28:46.178388+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.26\n2017-07-17T07:28:46.181666+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.27\n2017-07-17T07:28:46.184470+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.28\n2017-07-17T07:28:46.187460+00:00
ap-diagnostics[1222]: [INFO:report_fetcher.cc(159)] Starting fetch for 192.168.86.29\n2017-07-17T07:28:46.190281+00:00
broadcast=192.168.165.255 peer=\u003cunknown\u003e gateway=192.168.165.1\n2017-07-16T18:44:38.323369+00:00 INFO
broadcast=192.168.165.255 peer=\u003cunknown\u003e gateway=192.168.165.1\n2017-07-17T06:44:38.293098+00:00 INFO
buffet[581]: [INFO:shell_client.cc(496)] IPConfigProperty address = 192.168.165.9 at
channel", "newVersion": "0.0.0.0", "status": "idle", "blockedStations": {"stations": [], "child": {"lost": false}, "count
ChromeOSPolicy": {"updateCheckAllowed": END\n"}, {"path": "/var/log/webserverd/2017-07-17.log", "content": "192.168.86.22 --
D2C06860F068847E994DC698C8E94528F3104EDF7D04AA", "ipAddresses": ["192.168.86.22"], "lastSeenSecondsSinceEpoch": 0, "mdnsN
D3A-Q2Q-Q8B", "networkConfig": {"local_network": {"ip_address": "192.168.86.1", "netmask":
D6330272EAE838B7B1B1AB7F4A92E", "ipAddresses": ["192.168.86.21"], "lastSeenSecondsSinceEpoch": 0, "mdnsNames": [{"dp-535302
dc=yes\u0026ip=203.253.250.34\u0026ipbits=0\u0026mn=sn-i3beln7k\u0026ms=auh\u0026mt=1500256419\u0026mv=u\u00
dhcp(67) - SENT \n2017-07-17T04:41:30.132900+00:00 INFO dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.28
DHCPACK(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T11:11:46.869774+00:00 INFO kernel: [275264.642086]
DHCPACK(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T11:24:32.149875+00:00 INFO kernel: [276029.916534]
DHCPACK(br-lan) 192.168.86.21 a002dc000020 \n2017-07-17T05:16:14.290315+00:00 INFO kernel: [340332.058735]
DHCPACK(br-lan) 192.168.86.26 50f520000027 android.*****\n2017-07-17T06:00:45.750023+00:00 INFO
DHCPACK(br-lan) 192.168.86.27 d052a8000023 st-*****\n2017-07-16T20:39:24.561898+00:00 INFO
DHCPACK(br-lan) 192.168.86.28 109266000026 android.*****\n2017-07-17T06:11:55.609757+00:00 INFO
DHCPPOFFER(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T11:11:46.859574+00:00 INFO kernel: [275264.634667]
DHCPPOFFER(br-lan) 192.168.86.21 a002dc000020 \n2017-07-17T05:16:14.269776+00:00 INFO kernel: [340332.044510]
DHCPPOFFER(br-lan) 192.168.86.26 50f520000027 \n2017-07-17T06:00:45.747439+00:00 INFO dnsmasq-dhcp[2724]:
DHCPPOFFER(br-lan) 192.168.86.28 109266000026 \n2017-07-17T06:11:55.608522+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T11:11:46.867081+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T11:24:32.141409+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.21 a002dc000020 \n2017-07-17T05:16:14.283453+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.26 50f520000027 \n2017-07-17T06:00:45.747692+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.27 d052a8000023 \n2017-07-16T20:39:24.444539+00:00 INFO dnsmasq-dhcp[2724]:
DHCPREQUEST(br-lan) 192.168.86.28 109266000026 \n2017-07-17T06:11:55.609161+00:00 INFO dnsmasq-dhcp[2724]:
dnsmasq-dhcp[2724]: DHCPACK(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T08:23:44.696247+00:00 INFO
dnsmasq-dhcp[2724]: DHCPACK(br-lan) 192.168.86.24 1caf05000022
dnsmasq-dhcp[2724]: DHCPACK(br-lan) 192.168.86.25 b827eb000028 osmc\n2017-07-16T08:18:49.499776+00:00 INFO
dnsmasq-dhcp[2724]: DHCPACK(br-lan) 192.168.86.25 b827eb000028 osmc\n2017-07-16T20:18:50.112050+00:00 INFO
dnsmasq-dhcp[2724]: DHCPACK(br-lan) 192.168.86.25 b827eb000028 osmc\n2017-07-17T06:02:05.569635+00:00 INFO
dnsmasq-dhcp[2724]: DHCPNAK(br-lan) 192.168.165.23 2016d800001f wrong network\n2017-07-17T07:27:57.237318+00:00
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.165.23 2016d800001f \n2017-07-17T07:27:57.231199+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.21 a002dc000020 \n2017-07-16T08:23:44.583492+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.24 1caf05000022 \n2017-07-16T17:39:06.449977+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.24 1caf05000022 \n2017-07-17T06:00:47.191863+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.25 b827eb000028 \n2017-07-16T08:18:49.495580+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.25 b827eb000028 \n2017-07-16T20:18:50.005297+00:00 INFO
dnsmasq-dhcp[2724]: DHCPREQUEST(br-lan) 192.168.86.25 b827eb000028 \n2017-07-17T06:02:05.563285+00:00 INFO
```

7. (1 point) Please build up Betty's activity from these devices.

In the Amazon-Echo-Alexa-Web-Scrape, I was able to find the following as traces of Betty's Activity:

who yes
Today at 3:20 PM on simon's Echo
alex
Today at 3:20 PM on simon's Echo
turn off tv
Today at 3:20 PM on simon's Echo
alex
Today at 3:20 PM on simon's Echo
stop
Today at 3:13 PM on simon's Echo
alex
Today at 3:12 PM on simon's Echo
alex what do this to me stop we said we would what are you thinking
Today at 3:12 PM on simon's Echo
alex how could you do this what are the flooding
Today at 3:12 PM on simon's Echo
turn on pandora
Today at 3:06 PM on simon's Echo
alex
Today at 3:06 PM on simon's Echo
turn on tv
Today at 3:01 PM on simon's Echo
alex
Today at 3:01 PM on simon's Echo
alex stop
Today at 2:45 PM on simon's Echo
wake up
Today at 2:45 PM on simon's Echo
alex
Today at 2:45 PM on simon's Echo
Text not available. Click to play recording.
Today at 2:31 PM on simon's Echo

From the above we can trace that she was home around 2:31PM (UTC+9).

After filtering for Xiaomi apps by using the keyword “mi”, I was able to come across a few different db files, I looked through mistat.db and found the below:

New Database	Open Database	Write Changes	Revert Changes	Undo	Open Project	Save Project	Attach Database	Close Database
Database Structure	Browse Data	Edit Pragmas	Execute SQL					
Table: mistat_event								
id	category	ts	key	value	type	extra		
Filter	Filter	Filter	Filter	Filter	Filter	Filter		
1 3754	Event	1500276002646	Dashboard_ViewNum	1	event			
2 3755	mistat_session	1500276046304		1500275820512,1500276016276		WIFI		
3 3756	mistat_pv	1500276046323		.device.HMMiLiSettingActivity,.activ...				
4 3757	Event	1500276104636	Dashboard_StatusBar	1	event	{"Property":"BleOff"}		
5 3758	Event	1500276104636	BleOff	1	event			
6 3759	Event	1500280274725	Dashboard_ConnectionResults	1	event	{"Property":"Fail"}		

The above shows the events between betty's smartphone and the wearable device since the file was found on betty's phone.

Based on the above evidence and also filtering the files by "last written", I am able to tell that both Simon and Betty got home at 2 PM and then Betty was in the hall for the rest of the time interacting with Alexa.

For one of the timestamps, after converting to the unix epoch we get:

Convert epoch to human-readable date and vice versa

1500271649 **Timestamp to Human date** [\[batch convert\]](#)

Supports Unix timestamps in seconds, milliseconds, microseconds and nanoseconds.

Assuming that this timestamp is in **seconds**:

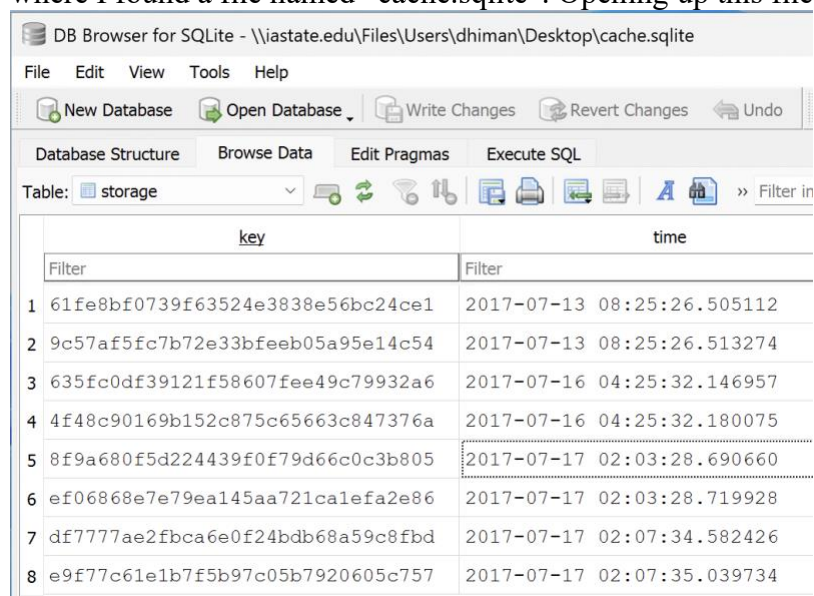
GMT : Mon, 17 Jul 2017 06:07:29 GMT

Your time zone : Mon Jul 17 2017 01:07:29 GMT-0500 (Central Daylight Time)

(For the timezone difference we add 9 hours to the GMT. The bottom is 15:07:29. 20 minutes before the crime is reported.)

8. (2 points) Simon claimed he arrived home at around 3 pm on 7th July 2017 and used a Bluetooth headset to watch TV, after which he found her on the ground when he entered the living room. Please verify his statement. Did he lie or not?

After searching using the keyword "Youtube", I was able to find multiple folders on the Raspberry Pi. Within them, I went into plugin.video.youtube and then into the folder kodion where I found a file named "cache.sqlite". Opening up this file I found the following details:



	key	time
Filter	Filter	
1	61fe8bf0739f63524e3838e56bc24ce1	2017-07-13 08:25:26.505112
2	9c57af5fc7b72e33bfeeb05a95e14c54	2017-07-13 08:25:26.513274
3	635fc0df39121f58607fee49c79932a6	2017-07-16 04:25:32.146957
4	4f48c90169b152c875c65663c847376a	2017-07-16 04:25:32.180075
5	8f9a680f5d224439f0f79d66c0c3b805	2017-07-17 02:03:28.690660
6	ef06868e7e79ea145aa721ca1efa2e86	2017-07-17 02:03:28.719928
7	df7777ae2fbca6e0f24bdb68a59c8fbd	2017-07-17 02:07:34.582426
8	e9f77c61e1b7f5b97c05b7920605c757	2017-07-17 02:07:35.039734

The above file shows the cache history for youtube.

After searching for the date 2017-07-17, I found a lot of files, I was specifically looking for files at time 01:00 AM CDT on 2017-07-17 to 02:00 AM CDT on 2017-07-17. This is because the ftk tool is showing me files in CDT and since the murder happened at 2017-07-17 15:31 (UTC+9), that would be around 01:00AM to 02:00AM CDT. Looking into the database folder with the time 01:20 AM CDT, I found the below file inside:

DB Browser for SQLite - \\iastate.edu\Files\Users\dhiman\Desktop\MyVideos107.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project

Database Structure Browse Data Edit Pragmas Execute SQL

Table: files Filter in any column

	idFile	idPath	strFilename	playCount	lastPlayed
	Filter	Filter	Filter	Filter	Filter
1	1	1	plugin://plugin.video.youtube/play/?...	1	2017-07-13 08:43:46
2	2	1	plugin://plugin.video.youtube/play/?...	1	2017-07-16 04:54:54
3	3	1	plugin://plugin.video.youtube/play/?...	1	2017-07-17 02:07:30
4	4	1	plugin://plugin.video.youtube/play/?...	NULL	2017-07-17 02:19:37

The above MyVideos108.db file proves that Simon was in fact watching videos in his room because there is a record of youtube playing at 2:07 PM (UTC+9) time and also another at 2:19 PM (UTC+9) time.

We can also tell from the files I found after filtering using date and time that the activities stopped from around 1:20 AM to 1:35 AM, which is probably the time that Simon decided to go out of his room and found his wife dead. Additionally these files are sorted using last written time so that the logs are accurate (when it shows 1:19 AM, it means that there was still some activity on Simon's devices at that time):

☐ 1430	profiles.xml	xml	07/17/17 01:19:36 AM (-5:00 Central Daylight Time)
☐ 1431	settings.xml	xml	07/17/17 01:19:36 AM (-5:00 Central Daylight Time)
☐ 1432	MyVideos107.db	db	07/17/17 01:19:37 AM (-5:00 Central Daylight Time)
☐ 1433	Database		07/17/17 01:19:37 AM (-5:00 Central Daylight Time)
☐ 1434	kodi.log	log	07/17/17 01:19:37 AM (-5:00 Central Daylight Time)
☐ 1435	entropy.dat	dat	07/17/17 01:34:18 AM (-5:00 Central Daylight Time)
☐ 1436	state-2017-07-17-14-02-44.bin	bin	07/17/17 01:34:35 AM (-5:00 Central Daylight Time)
☐ 1437	procstats		07/17/17 01:34:35 AM (-5:00 Central Daylight Time)

Based on all the above proof, I can conclude that Simon did not kill his wife because there is proof of him being in his room and watching youtube.